

applied to many different 2D materials to study their intrinsic defects[205] and most recently *in situ* microscopy of liquid systems[196, 206].

2.1.8 Characterisation of van der Waals' heterostructures

Currently, the primary method for characterising a van der Waals' heterostructure is the electronic transport measurements or the measurement of a physical phenomenon. For the most part this is due to van der Waals' heterostructures being relatively new and so standards in quality have not been established. Characterisation by this method relies on the sensitivity of the phenomenon being measured, for example simple electron mobility measurements are less sensitive to structural defects and impurities than observing charge density wave propagation. Until relatively recently any evidence of a device working satisfied the burden of proof for its purpose (i.e. publication). This has now changed as the gold rush matures, demanding standards in device reproducibility and measurements to aid comparison between studies. Here we will briefly review some common characterisation techniques for isolated flakes and heterostructures outwith electron microscopy.

The most common characterisation techniques for probing device structure and quality are atomic force microscopy (AFM) and Raman spectroscopy. These are often used not only for heterostructures but on the isolated 2D material flakes prior to assembly. In AFM an atomically sharp tungsten needle attached to a cantilever is rastered over the surface of a specimen. As the tip interacts with the specimen surface, the cantilever is deflected. This deflection is detected by a laser reflecting off the back of the cantilever and onto a sensitive photodiode. As the cantilever is deflected in different

directions, so too is the laser beam on the diode and so the forces the tip experiences are translated into a digital signal.[207] There are a myriad of 'modes' that the AFM can operate under, including constant force mode and tapping mode, which serve to selectively detect different surface properties of the material. The most common properties measured in 2D materials by AFM are height changes[143, 208, 209, 210], surface roughness[22, 143], and deformation properties[211, 212, 213, 214]. AFM is a hugely popular technique due to its versatility and reliable non-destructive characterisation of surfaces at atomic resolution. It is, however, very much limited to surface properties, is largely insensitive to chemical composition and can sometimes interfere with the specimen e.g. pushing contamination ahead of the tip, thereby cleaning a surface before measurement. AFM (as with all characterisation techniques) is best used in conjunction with other methods to fully characterise a material or heterostructure.

Raman spectroscopy has been fundamental in the characterisation of low dimensional carbon systems[154, 215]. The relationship between graphene and Raman spectroscopy is particularly special thanks to the wealth of information that can be harvested from a careful analysis of the few resonance peaks that populate its characteristic spectrum. Raman spectroscopy uses monochromatic light from a laser to excite vibrational and rotational states in the specimen. Each of these states is associated with a frequency shift in the reflected light as a result of inelastic scattering. The spectrum that results from this interaction can be routinely used to identify chemical compounds, but for the skilled spectroscopist a variety of material properties can be investigated with careful experimental design and signal analysis. In

pristine graphene there are two dominant peaks named the G band ($\sim 1580 \text{ cm}^{-1}$) and the G' band ($\sim 2700 \text{ cm}^{-1}$). The latter is also sometimes known as the “2D” band, however this can become confusing in terms of its origin and when used to characterise two dimensional materials[216]. For disordered or defective sites in graphene the D band is detected at about half the G' band wavenumber ($\sim 1350 \text{ cm}^{-1}$). The ratios between these peaks have been used to quantify the quality of graphene as a result of fabrication[142, 217] and in irradiation experiments where defects are generated in the graphene lattice [218, 219]. The peaks themselves, however, can also be meticulously analysed to reveal the thickness of graphene with single-layer precision[154], the stacking configuration of the basal planes [220, 221], doping[222] and changes in lattice strain[24]. Raman spectroscopy is now also used routinely for other 2D materials, including hBN [223, 224] and TMDCs [225, 226, 69, 67, 227], to probe similar characteristics.

Raman spectroscopy, unlike AFM, is not purely a surface technique. The light interacts with a volume of material below the incident surface and has a penetration depth of 50 - 100 nm in graphite[228]. Most Raman spectrometers are capable of 1 μm spatial resolution, so whilst complementary to many other techniques it is often difficult to directly assign nanoscale phenomena with Raman signals diffuse over micrometer areas. Nevertheless, Raman spectroscopy is invaluable in the field of 2D materials as it is the only techniques which offers high through-put, non-destructive characterisation of materials.

2D materials stacked in a heterostructure present a problematic characterisation challenge: How do we probe the properties of two interfacing crystals

and the nature of the interface? Is the interface free from contaminants and which bonding regime are the crystals using to adhere? AFM cannot answer many of these questions, although height and roughness maps can help to infer device composition and quality[229, 230]. Raman spectroscopy can be used to investigate buried interfaces thanks to the extended penetration depth of light in these materials[228]. This works for well characterised materials like MoS₂ and graphene, where the peak positions are well known. Deviations in peak positions and widths can then be attributed to different phenomena, for example bands arising from in-plane vibrations carry information about strain in the lattice whilst out of plane vibrations correspond to the quality of the interfacial contact[226]. This method works well if all the superimposed spectra from the heterostructure components can be easily identified, however as heterostructures include more components of different materials (all at different depths) the peaks in the Raman spectra will quickly become very difficult to assign. Nevertheless, Raman spectroscopy of van der Waals heterostructures is still relatively niche and the application of this technique will almost certainly grow in popularity as it has for isolated flakes of 2D materials.

A similar but lesser known technique which provides useful information about heterostructures is optical second harmonic generation (SHG). SHG uses a non-linear optical interaction of light with a crystal to convert two low energy photons into a single high energy photon, in which is encoded information about the interface. SHG is only possible from noncentrosymmetric crystals and so cannot be used to probe the properties of graphene, which perhaps explains its poor popularity. Nevertheless, monolayer and odd-numbered layer

hBN and TMDCs are noncentrosymmetric and so give rise to strong SHG in an optical probe[231, 232, 233, 234]. This technique could prove prevalent in the future of van der Waals device characterisation as interfaces at TMDCs can be characterised nondestructively, with stacking orientation, crystal polarity and domain boundaries mapped from SHG[235]. The biggest advantage this technique presents is the nondestructive determination of crystallographic orientation for individual crystals by purely optical methods[236], however it is only useful for odd-numbered layered, noncentrosymmetrically stacked crystals.

The first destructive technique we will explore is time of flight secondary ion mass spectrometry (TOF-SIMS). Secondary ion mass spectrometry (SIMS) uses energetic ions to sputter material (secondary ions) from the specimen surface in ultra high vacuum. The secondary ions are collected using an electrostatic lens close to the specimen and sent to the analyser. There are a few different analysers possible, however the most useful is a “drift tube” as all ion species can be analysed simultaneously. In the drift tube the ions are accelerated by a known electric field and their mass-charge ratio is determined by their time of flight (TOF) through the tube. This analysis method relies on pulsing the primary beam to instantaneously generate secondary ions for an accurate determination of species. The primary beam can also be rastered across the surface of the specimen to generate a map of mass spectra with a lateral resolution of ~ 200 nm and sub-nanometer depth resolution. This technique is not common to the field of 2D materials yet, however preliminary studies of heterostructures show interesting results: It is the only technique which has detected hydrogen between atom-

ically thin sheets, showing the sensitivity of the mass analysis and depth resolution[237]. It has also demonstrated graphene functionalisation below the sensitivity of Raman spectroscopy[238]. When applied to complex heterostructures, TOF-SIMS can not only identify the 2D materials at different depths, it can identify contaminant chemical residues between layers from the secondary ion species[229]. To do this accurately, however, AFM measurements are required prior to TOF-SIMS to deconvolve intrinsic crystal roughness from that induced by ion collisions. This is part of a more complex “depth resolution function” which separates different contributions to the signal to yield the correct placement of species with depth and explains why the interfaces are not detected as atomically sharp. Nevertheless, this is a powerful (though specialist) method for detecting contaminant species dormant in heterostructures which may not be detectable by other methods.

Finally we will discuss density functional theory (DFT) calculations of heterostructures. Although this is not a characterisation method, it is the go-to technique for comparing the high resolution structural information obtained by (S)TEM techniques (see section 2.2) and is used in two of the papers comprising the final chapters. In 1998 Walter Kohn was awarded the Nobel Prize in Chemistry for his deduction that the properties of a chemical system can be predicted if the density distribution of electrons in their ground state is known[239]. Previous to this, every electron in a model structure had to be accounted for which was hugely computationally expensive and greatly limited the size of the systems that could be studied. DFT lifted this burden by approximating the wavefunctions of electrons into “density functionals”, which the theory aims to solve for by finding the overall min-

imum energy of the system[240, 241]. This allowed the structures of multi-bodied systems (i.e. comprising many nuclei and many more electrons) to be optimised with tolerable computational cost, ultimately proving it to be the most popular theoretical approach for problems in materials science and chemistry[242]. There are different “flavours” of DFT methods which use various approximations to help describe different types of systems where certain interactions are prevalent. The motivation to model heterostructures of graphene and other 2D materials comes from the difficulty in physically realising these complex structures; it is far more efficient to iterate through different architectures, substrates, twist angles and combinations of 2D materials theoretically than it is to fabricate and characterise the myriad possibilities in the lab[243]. DFT is currently used to predict theoretical structures and properties which experiment is yet to realise, and also to explain experimental observations using simulations[244, 41, 245, 246, 247]. Van der Waals’ materials are an odd class of materials in that they simultaneously have strong short-range chemical bonding in two cartesian directions and weak long-range bonding normal to these. Accurately modelling both bonding types has proved very difficult in DFT as dispersion forces have typically been approximated using corrections[248]. The most promising method to address this is the Becke88 van der Waals functional (optB88-vdW)[248] which has shown excellent agreement with experimental data for graphite and hBN crystal structures[249]. Nevertheless there is no particular consensus in the community as to which method is best as even optB88-vdW translates poorly to other van der Waals systems and different crystal thicknesses[250]. As such, other DFT corrections such as optPBE-vdW, vdW-DF2 and DFT-D2 are also still frequently used in the community. This is also in part due

to a dearth in accurate experimental data for these exotic systems, as the methods described in the rest of this section do not provide comparable data. Cross sectional (S)TEM is the only technique thus far to yield atomically and chemically accurate data of van der Waals' heterostructures comparable to DFT results.

2.2 (S)TEM imaging of 2D materials

2.2.1 TEM and STEM

Since TEM was first demonstrated by Ruska and Knoll in 1932[251] it has underpinned advances in materials science ever since. Its primary advantage comes in the form of imaging resolution, although the charged nature of the electrons and their different interactions with the specimen are what make TEM stand out as a technique. The wavelength of visible light is 400 - 700 nm (from blue to red) and, due to Abbe's diffraction limit, the highest resolution we can hope for with conventional visible-light microscopy is roughly in the range of 300 nm. Whilst this resolution is fine for imaging many structures in biology (e.g. cell structures), to look at atoms in a crystal structure we require resolution on the order of Ångströms. This is best achieved using high energy electrons which have wavelengths close to 1 pm, however aberrations in the electron microscope system quickly become the limiting factor for resolution[252]. Advances in lens aberration correction, and now monochromation of the electron beam, underpin the growing popularity and powerful application of TEM to a range of problems in materials science[253, 254].

There are two ways of imaging a specimen with transmitted electrons: conventional TEM (CTEM) and scanning TEM (STEM). In both, electrons are pulled from the tip of a source (typically either a specially grown LaB₆ crystal or an incredibly sharp tungsten needle) by a set up not dissimilar to that in an old cathode ray tube television, except with the additional complication of operating at 10 - 300 kV. The electrons are accelerated through this

huge potential and exit the electron gun. It should be noted that not only do the electrons exhibit both particle- and wave-like character, they are now also relativistic particles. On leaving the gun, the beam of electrons is focussed and refined by a series of apertures and lenses which comprise the rest of the condenser system. Here the two techniques diverge: In CTEM the condenser lens forms a broad, near-parallel electron beam which passes through the specimen and is collected by the objective lens to form an image on the other side. The projector system then magnifies this image onto a phosphor screen for viewing or a charge coupled device (CCD) camera for digital readout. In STEM the condenser lens (or probe forming lens) focuses the beam into a sharp probe which is then rastered across an area of the specimen, where each probe position corresponds to a pixel in a digital image. For each pixel, detectors collect the electrons in the post-specimen beam which results in a signal which is amplified and used by a computer to digitally reconstruct the scan area. Each detector collects electrons across different angular ranges, which correspond to different scattering phenomena (the angular ranges can be changed by the microscope user for different materials). The typical detectors used in commercial STEM systems are bright field (BF) and high angle annular dark field (HAADF) detectors. A separate image of the area scanned can therefore be generated for each detector, each holding different information about the same area. The pixel-by-pixel acquisition used in STEM is a major advantage over CTEM as not only can many different signals be recorded simultaneously, but these can be mapped across an area to give complementary information on sub-nanometre regions of material. Other differences to note are that in CTEM the resolution is limited by aberrations in the image forming objective lens and in STEM

the resolution is limited by the beam diameter achieved by the condenser lens, which is also ultimately limited by aberrations. Aberration correction is slightly simpler for STEM than for CTEM, as the former does not use an image-forming lens, making high resolution aberration corrected STEM imaging a popular and powerful tool in materials science[255].

All the images and spectrum datasets in this work have been acquired using an aberration corrected STEM using 200 kV electrons. Despite great advantages of TEM, there are some limitations to this technique. The first is that, compared to a lot of other characterisation methods which typically populate a materials science department (e.g. X-ray diffraction, all light spectroscopy techniques, light microscopy, thermal gravimetry techniques, etc), TEM can realistically only image a very small fraction of a specimen and it is estimated that the cumulative total volume of material imaged in TEM is only 1 m^3 in volume[251] (2009 estimate). This means that the area analysed must be representative of the specimen studied and, if possible, compared to other techniques which sample higher volumes. The second problem is that the specimen has to be able to withstand high vacuum conditions, which almost immediately limits our range of useful samples to solid systems. Nevertheless, specialist environmental TEM instruments exist where the vacuum around the specimen is greatly diminished to simulate low pressure chemical environments[256], and more recently liquid cells have become commercially available which allow conventional (S)TEM instruments to image nanoparticles in liquid and gas environments [257, 258]. The specimen has to be thin enough for the electrons to pass through ('transmission') and so that multiple scattering events do not dominate the resolution. This means that

specimen preparation is crucial if high resolution imaging is needed. As we will discuss in section 3.2.12, specimen thickness almost certainly limits the resolution of the STEM images in this work. The final limitation is the specimen's response to the electron beam (see section 2.2.5). This is a particular problem for unstable materials which can undergo sputtering or radiolytic damage, chemical reactions or particle aggregation under the electron beam[259, 260, 261]. In the next section we look at the mechanisms underlying BF and HAADF STEM images as these form the basis of analysis in this work.

2.2.2 STEM imaging

As described in the previous section, a STEM image is formed from the pixel-by-pixel digital readout from a detector collecting post-specimen electrons within a defined angular range. The reason for this is that if an incident electron interacts with either the electron cloud or nucleus of a specimen atom, then the electron will be forced to deviate from its path (i.e. scattered) to different degrees. We can therefore broadly categorise the post-specimen beam into the unscattered (or direct) beam, coherently scattered electrons, and incoherently scattered electrons. If we understand the contrast mechanism used to form an image from a detector we can attribute physical meaning to it and even perform quantitative measurements.

The bright field detector is placed on the optical axis of the microscope, collecting the direct beam and electrons scattered very weakly by the specimen. Phase contrast, diffraction contrast and mass-thickness contrast all contribute to a STEM bright field image and we will briefly describe each

of these in turn. Phase contrast arises from interference between the direct beam and the first Bragg diffracted beams from the specimen at the detector. The phase contrast of an image is determined not only by the arrangement of atoms in a specimen but by the microscope operating conditions. We can derive the phase contrast dependence by considering the intensity generated at a detector I_{coh} by coherently interfering electrons passing through a point r in a specimen with exit wave $\psi(r)$ and where $P(r)$ is the complex point spread function of the electron microscope[262].

$$I_{coh}(r) = |\psi(r) \otimes P(r)|^2 \quad (1)$$

This is the general case for coherent imaging, used in both BF STEM and high resolution TEM (HRTEM). If we assume that the specimen is a “weak phase object”, i.e. the specimen is thin enough so that only a small phase shift occurs as the electron wavefunction passes through the object, it can be shown that the intensity of coherent scattering varies with spatial frequency g as $\sin\chi(g)$. This is the phase contrast transfer function (PCTF) and must first be calculated if a quantitative interpretation of a pure phase contrast image (e.g. HRTEM) is desired, as even changes in defocus will lead to contrast inversion under certain conditions. In BF STEM imaging there is an additional contribution from diffraction contrast, however it is usually a very minor contribution due to the small collection angle of this detector, typically showing up as bend contours in thin foils [251]. The final contribution is mass-thickness contrast which is simply the contrast generated by the increased scattering associated with a denser or thicker region of specimen. BF STEM imaging is a useful complement to bright field imaging as the

combination of all these contrast mechanisms can generate very sharp images which can aid interpretation, however they cannot provide quantitative information without careful modelling of microscope conditions and image simulation.

HAADF STEM imaging uses a large radius annular detector to collect electrons elastically scattered to high angles, typically through interaction with the nuclei of atoms. In this regime the scattering cross section σ through an angle ϑ can be described by relativistic Rutherford scattering[263]

$$\frac{d\sigma}{d\Omega} = \frac{4\gamma^2 Z^2}{a_0^2 (2k \sin(\frac{\vartheta}{2}))^4} \quad (2)$$

with the relativistic factor for the electron γ given by

$$\gamma = [1 - \frac{v^2}{c^2}]^{-\frac{1}{2}} \quad (3)$$

and

$$k = \frac{2\pi}{\lambda} \quad (4)$$

where Z is the atomic number, a_0 the Bohr radius, v the electron velocity, c the speed of light, and k the magnitude of the wave vector of the electron, which is related to the wavelength of the electron λ by equation 4. The annular geometry of the detector means it selectively collects scattered electrons, whilst the direct beam and the bright field cone passes through the centre of the detector. This makes HAADF a dark field technique, where the positions of atoms generate a high intensity signal ('bright'). This differs to

BF imaging as atoms can be either be dark or bright pixels depending on the microscope conditions. HAADF images are directly interpretable as high angle scattering disrupts the coherent nature of the electron wavefunction. The intensity at the detector I_{incoh} for a probe at position r can be expressed as

$$I_{\text{incoh}}(r) = |\psi(r)|^2 \otimes |P(r)|^2 \quad (5)$$

If we take an annular detector to collect incoherently scattered electrons, it can be shown that by optimising the inner angle radius to exclude the contribution of diffraction spots to the signal (>50 mrad), the intensity I at the detector becomes

$$I \propto \sigma^2 Z^2 \quad (6)$$

where σ is the mean square displacement of the atom and Z is the atomic number[264]. So not only is the contrast of the HAADF image directly interpretable as a dark field signal, the intensity is proportional to the mean square displacement of the atom (which is temperature dependent) and the square of the atomic number. This is what is known as Z^2 contrast in the electron microscopy community, as (for a suitably thin, flat specimen) chemical information can be quantitatively deduced from the images directly e.g. identifying the composition of doped atomic columns [265, 266] or counting the number of atoms in a column to reconstruct a nanoparticle [267, 268]. However due to relativistic screening in experiment the contrast actually varies Z^ζ , with ζ in the range of 1.5 - 2.0.

In this work the quantitative nature of HAADF contrast has not been fully exploited for chemical information, however the dark field and directly interpretable information present in the technique is used to accurately determine the positions of atomic fringes, leading to quantitative analysis of atomic scale structures and geometries. We will now explore complementary techniques which further extend the chemical analysis of the STEM instrument.

2.2.3 Analytical spectroscopic methods: EDXS

When incident electrons inelastically scatter through an electron cloud, if the accelerating voltage is high enough, core electrons of the atom can be kicked out to leave a hole. The hole is quickly filled by an electron from a higher energy orbital, which requires the emission of an X-ray photon. The photon will have the energy difference between the electrons initial state and when it has filled the hole. This photon energy is characteristic of the atom it radiates from, as the core orbitals are unscreened from the nucleus charge and so give unique photon energies for different elements. We can use this as a fingerprint to determine the composition of a specimen.[251] Unlike many TEM techniques, EDXS requires a thick specimen to generate enough counts so that the signal is detectable above the background X-rays (named the bremsstrahlung and arising from the deceleration of electrons through the specimen). Furthermore, X-rays from light elements have very low photon energies and are very difficult to get good statistics with EDXS and so are usually consigned to detection by EELS if possible (see section 2.2.4).

EDXS operates differently in CTEM and STEM. In CTEM the entire area

illuminated by the broad beam contributes to one spectrum, from which the composition of the specimen is inferred. In STEM, a spectrum can be acquired from each probe position. This is known as Spectrum Imaging (SI) and is very powerful as the local composition of nanostructures can be determined by mapping characteristic peak intensities across the image. Furthermore EDXS SI can be done in tandem with HAADF STEM imaging and BF imaging or EELS SI to gather multiple signals simultaneously for the same region of specimen. As X-ray emission occurs in all directions EDXS is not limited to transmission techniques and is also used ubiquitously in scanning electron microscope systems.

2.2.4 Analytical spectroscopic methods: EELS

Electron energy loss spectroscopy (EELS) extends the use of the electrons in the microscope from purely imaging to giving us extensive elemental and chemical information about the specimen.[269] Incident electrons can experience a multitude of inelastic scattering events on passing through the specimen, resulting in them losing the initial energy gained in the electron gun. We can separate out these electrons by their energy loss using an electromagnetic prism to form a spectrum. The loss energies associated with different scattering phenomena are all present in the spectrum. As we require the direct beam and weakly scattered electrons about this the bright field detector is removed from the optical axis when using the EELS detector.

The low loss region (0 - 50 eV) of the spectrum contains the zero loss peak, which contains all elastically scattered electrons from the specimen, and plasmon peak(s). If the specimen is thick (e.g. > 90 nm for silicon) two or more

plasmon peaks will be present and the specimen will not be suitable for EELS analysis (although EDXS count rate will not be problem, see section 2.2.3) For suitably thin specimen, where only one scattering event is present, there will only be a single plasmon peak. Taking the ratio of this peak and the zero loss peak allows you to calculate the mean free path of an electron through the specimen. This can be converted into specimen thickness using the log-ratio method[270, 271]. It is also useful to record the low loss region for correction of the high loss region as not only does the energy dispersion shift during acquisition but the core loss EELS edges are convolved with the high intensity peaks in the loss region and should ideally be deconvolved for quantitative analysis. Recording both low and high loss regions is known as dual-EELS.

The high loss region is far less intense than the low loss, however it contains the most useful signals coming from the specimen. A core-loss edge corresponds to a scattering process where a core electron has been promoted to an unoccupied state just above the fermi level.[263] The position of the edge can be used to identify the element and the fine structure of the edge (i.e. small peaks after the edge onset) can provide information on the chemical environment of the species[272]. For excitations with low cross sections the signal can be very low, especially for higher energy excitations with a gradual edge onset. EELS is particularly useful for identifying light elements and is highly complementary to EDXS.

2.2.5 Electron beam induced effects

If the microscope and specimen are both optimised, the changes induced in the specimen by the electron beam are the next thing to limit the quality of the image or spectrum image in STEM.

Whatever specimen you are looking at, the most common problem is contamination build up during imaging. This has been an issue in TEM for a very long time[259], however the situation has improved with advances in vacuum systems. Nevertheless, it is still very common in all electron microscopy methods to experience a gradual change in image contrast with increased electron fluence, as mobile carbon contamination moves across the surface of the specimen into the highly charged area under the electron beam. Strategies to combat this are cooling the specimen, 'beam showering' and careful oxidative plasma cleaning of the specimen[261]. For extremely sensitive specimen, heating by the electron beam can also induce changes, however these are estimated to be on the order of 1 °C for a suitably thin sample in CTEM and STEM[260]. A far bigger problem is direct damage induced by the electron beam. Generally this takes the form of knock-on damage or radiolytic damage. Knock-on damage is when an incident electron collides with an atomic nucleus, either irreversibly displacing it from its original position or ejecting it from the specimen. The cross sections for knock-on damage for light elements, such as carbon, boron and nitrogen, are significant even in covalently bound systems[273] and the electron beam energy should be reduced so that incident electrons do not breach the sputtering threshold of the species [200]. Radiolysis is damage initiated by ionising an atom in the specimen, and where the hole is not filled within the timescale of phonons.

This results in the weakened bond being torn apart by lattice vibrations, which in turn leads to a release of secondary electrons and generating a local cascade of radiolytic damage[274]. Radiolysis is mostly confined to semiconductors and insulators, as the dissociated electron cloud in metals almost instantaneously fills any holes generated. Radiolytic damage can be reduced somewhat by cooling the specimen to reduce thermal vibrations, making the specimen thinner, or increasing the electron beam energy[251]. The final electron beam effect we will discuss is electrostatic charging. A strong electrostatic field can be induced in thin, insulating specimen subject to a STEM probe. This arises from the generation of charged species and ejection of secondary electrons from the specimen and has been shown to have different polarity in different regions of a wedge specimen[275]. Anion and cation species are made mobile by the electrostatic field and can cause breakdown of the specimen structure[276].

All these affects were encountered in this work due to the beam sensitive nature of many of the components of atomically thin van der Waals' materials, many of which were bulk hBN dielectric. Characterisation of the extent of beam damage from FIB and STEM irradiation is demonstrated in section 3.2.12.

2.2.6 Advanced signal processing techniques for image and spectrum datasets

To get quantitative information from raw TEM data, some data processing is usually required to . We will briefly cover some of the techniques used in this work.

A common technique in TEM imaging is to convert an image to a fast fourier transform (FFT), filtering or masking in the frequency domain, and using an inverse fast fourier transform to recover the desired data in the image. This is usually done with the intention of removing low frequency noise from the image[251] or subtracting the information from a specific lattice in order to better interpret the image (usually done in conjunction with image simulation)[277].

For direct processing of images some common techniques are gaussian blur, for smoothing spatial fluctuations; thresholding to generate a binary image to isolate specific features; edge detection; Hough transforms, for detecting straight lines; and skeletonisation, which reduces a multi-pixel structure down to a single thread of pixels[278]. The last technique is very useful for quantitative analysis in TEM images as it allows the position of a lattice fringe to be expressed as an absolute pixel coordinate, and is used by the 'PyroMaN' software[279] to extract quantitative information from HRTEM lattice fringes[280, 281]. Many of these processing techniques have been used in this work on HAADF STEM images, however each one is applied image-wise with close supervision of the outcome for each technique, and not as a black box algorithm applied to a stack of images (see section 3.2.11).

Multidimensional-data analysis is also possible for both STEM images and spectrum images[282]. A dataset only requires two fundamental dimensions: the "signal" axis, which plots the amplitude of some signal or measurement; and a "navigation" axis, which tells us over what metric the signal fluctuates. More than one type of axis can be assigned e.g. a greyscale TEM image has two navigation axes and one signal axis, a simple spectrum has

only one of each, and a spectrum image has three navigation axes (two for spatial probe position, one for energy) and one signal axis (X-ray or electron counts). By assigning axes correctly we can apply multidimensional-data analysis effectively to any dataset and extract the components of the signal we desire and map them across the navigation axes. A common branch of multidimensional-data analysis is blind source separation which attempts to disentangle a set of variables which outpopulate any relationships between them[283]. Machine learning algorithms such as Principle Component Analysis (PCA) can distinguish parts of a complex signal based on their variance and, with some human input, be used to remove noise and identify signal components with physical meaning[284]. PCA has been successfully applied to multidimensional datasets in microscopy, such as EELS and EDXS SIs[285, 286, 287].

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